

Conducting an a priori power analysis

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Load packages:

```
library(MBESS)      # account for uncertainty in effect sizes
library(dplyr)      # tidy data
library(purrr)      # iterations
library(effectsize) # calculate effect sizes
library(broom)      # tidy statistical objects
library(faux)       # simulate data for factorial designs
library(pwr)        # power analysis for basic designs (eg., t-test)
library(Superpower) # power analysis for factorial design
library(TOSTER)     # equivalence tests
library(pwr2ppl)    # power analysis for basic designs and logistic regression
library(emmeans)    # marginal means and post hoc comparisons
```

Account for uncertainty and bias when using an published effect size estimate

Suppose we are interested in answering the following research question (RQ):

- *Does the consumption of creatine improve power output during a cycling time trial compared to a placebo?*

We rely on a published study and assume that the true effect size is similar to the one reported. However, publication bias may inflate reported effects, and many studies are, on average, underpowered.

As a result, there is **uncertainty about the true effect size**, which should be explicitly accounted for when planning the study.

To address this, we can use the MBESS package to incorporate **uncertainty in effect size estimates** into the power analysis, leading to more robust sample size justification.

```
res <- ci.smd(
  smd = 0.5,
  n.1 = 20,
  n.2 = 20,
  conf.level = 0.60
)
```

Conduct a power analysis using the lower bound of the 60% CI:

```
pwr.t.test(
  d = res$Lower.Conf.Limit.smd,
  power = 0.9,
  sig.level = 0.05,
  alternative = "greater"
)
```

Two-sample t test power calculation

```
      n = 335.2629
      d = 0.2262542
sig.level = 0.05
  power = 0.9
alternative = greater
```

NOTE: n is number in *each* group

ANOVA

In sports and exercise science, **factorial designs** are common. These designs can be implemented as **between-subjects** (each participant is assigned to one condition), **within-subjects** (each participant experiences all conditions), or **mixed designs** (one factor is between-subjects and the other within-subjects). This structure enables researchers to examine both the **main effects** of each factor and their **interaction effect**.

Superpower

Superpower (Caldwell et al. 2022) is an *R* package (also available as [Shiny app](#)) to conduct power analyses. One of the main benefits of **Superpower** is that it does not require a standardized effect size, in contrast to tools such as G*Power or **pwr**.

Between-subject ANOVA

Suppose we are interested in testing these three RQs:

- *Does consuming chocolate affect power output during a time trial compared to water?*
- *Does consuming dates affect power output during a time trial compared to water?*
- *Does consuming raisins affect power output during a time trial compared to water?*

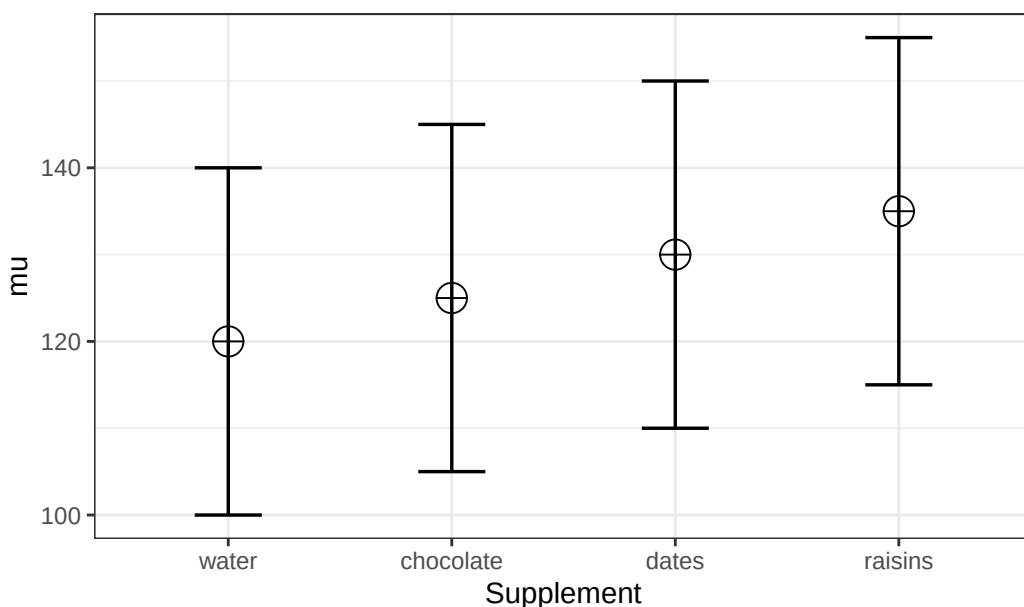
Note

Note these are three different RQs (hypotheses) so we do not need to adjust for multiple comparisons

We first begin by setting the parameters to simulate data based on our study design using `ANOVA_design()`:

```
design_result <- ANOVA_design(  
  design = "4b",  
  n = 80,  
  mu = c(120, 125, 130, 135),  
  sd = 20, # common standard deviation  
  labelnames = c("Supplement",  
                 "water", "chocolate", "dates", "raisins"),  
  plot = TRUE  
)
```

Means for each condition in the design



We then use `ANOVA_exact` to conduct the computation:

```
res <- ANOVA_exact(
  design_result,
  alpha_level = 0.05,
  emm = TRUE
)
```

Power and Effect sizes for ANOVA tests

	power	partial_eta_squared	cohen_f	non centrality
Supplement	99.2696	0.0733	0.2813	25

Power and Effect sizes for pairwise comparisons (t-tests)

	power	effect_size
p_Supplement_water_Supplement_chocolate	34.91	0.25
p_Supplement_water_Supplement_dates	88.16	0.50
p_Supplement_water_Supplement_raisins	99.71	0.75
p_Supplement_chocolate_Supplement_dates	34.91	0.25
p_Supplement_chocolate_Supplement_raisins	88.16	0.50
p_Supplement_dates_Supplement_raisins	34.91	0.25

Power and Effect sizes for estimated marginal means

	contrast	power	partial_eta_squared	cohen_f
--	----------	-------	---------------------	---------

1	Supplement_chocolate - Supplement_dates	35.08	0.0078	0.0889
2	Supplement_chocolate - Supplement_raisins	88.35	0.0307	0.1779
3	Supplement_chocolate - Supplement_water	35.08	0.0078	0.0889
4	Supplement_dates - Supplement_raisins	35.08	0.0078	0.0889
5	Supplement_dates - Supplement_water	88.35	0.0307	0.1779
6	Supplement_raisins - Supplement_water	99.72	0.0665	0.2668
	non centrality			
1	2.5			
2	10.0			
3	2.5			
4	2.5			
5	10.0			
6	22.5			

```
res$aov_result
```

Anova Table (Type 3 tests)

Response: y

	Effect	df	MSE	F	pes	p.value
1	Supplement	3, 316	400.00	8.33 ***	.073	<.001

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

Power for the effect of Supplement:

```
res$main_results
```

	power	partial_eta_squared	cohen_f	non centrality
Supplement	99.26959	0.07331378	0.281272	25

Power for each of the post hoc comparisons:

```
res$pc_results
```

	power	effect_size
p_Supplement_water_Supplement_chocolate	34.90542	0.25
p_Supplement_water_Supplement_dates	88.16025	0.50
p_Supplement_water_Supplement_raisins	99.70604	0.75
p_Supplement_chocolate_Supplement_dates	34.90542	0.25
p_Supplement_chocolate_Supplement_raisins	88.16025	0.50
p_Supplement_dates_Supplement_raisins	34.90542	0.25

Because we are not interested in all possible comparisons, we can use specific contrasts with the `emmeans` package. Since our focus is on comparing each supplement to water, we can set water as the reference group and perform targeted comparisons accordingly.

```
(comparisons <- emmeans(
  res$emmeans$emmeans,
  specs = trt.vs.ctrl ~ Supplement,
  ref = 4,
))
```

```
$emmeans
  Supplement      emmean    SE  df lower.CL upper.CL
Supplement_chocolate    125 2.24 316      121      129
Supplement_dates        130 2.24 316      126      134
Supplement_raisins       135 2.24 316      131      139
Supplement_water        120 2.24 316      116      124
```

Confidence level used: 0.95

```
$contrasts
  contrast                                estimate    SE  df t.ratio p.value
Supplement_chocolate - Supplement_water          5 3.16 316   1.581  0.2724
Supplement_dates - Supplement_water             10 3.16 316   3.162  0.0050
Supplement_raisins - Supplement_water            15 3.16 316   4.743 <0.0001
```

P value adjustment: dunnett method for 3 tests

Power for the comparisons of interest:

```
emmeans_power(comparisons$contrasts)
```

```

      contrast      power partial_eta_squared
1 Supplement_chocolate - Supplement_water 35.08293      0.007849294
2 Supplement_dates - Supplement_water 88.35045      0.030674847
3 Supplement_raisins - Supplement_water 99.71885      0.066469719
  cohen_f non centrality
1 0.08894601      2.5
2 0.17789202     10.0
3 0.26683803     22.5
```

Now suppose we are interested the following RQ:

- *Does the consumption of chocolate milk affect power output during a time trial compared to placebo?*
- *Does the consumption of chocolate milk affect power output during a time trial compared to dates?*

```
design_study <- ANOVA_design(
  design = "3b",
  n = 20,
  mu = c(120, 135, 130),
  sd = 20, # common standard deviation
  labelnames = c("Supplement",
                 "placebo", "chocolatemilk", "dates"),
  plot = TRUE
)
```

```
res <- ANOVA_exact(
  design_study,
  alpha_level = 0.05,
  emm = TRUE
)
```

Power and Effect sizes for ANOVA tests

	power	partial_eta_squared	cohen_f	non centrality
Supplement	54.7619	0.0928	0.3199	5.8333

Power and Effect sizes for pairwise comparisons (t-tests)

	power	effect_size
p_Supplement_placebo_Supplement_chocolatemilk	63.74	0.75
p_Supplement_placebo_Supplement_dates	33.79	0.50
p_Supplement_chocolatemilk_Supplement_dates	12.03	-0.25

Power and Effect sizes for estimated marginal means

	contrast	power	partial_eta_squared
1	Supplement_chocolatemilk - Supplement_dates	12.16	0.0108
2	Supplement_chocolatemilk - Supplement_placebo	64.49	0.0898
3	Supplement_dates - Supplement_placebo	34.28	0.0420

	cohen_f	non centrality
1	0.1047	0.625
2	0.3141	5.625
3	0.2094	2.500

```
comparisons <- emmeans(
  res$emmeans$emmeans,
  specs = ~ Supplement
)
```

```
contrast <- contrast(
  comparisons,
  list(rq1 = c(1, 0, -1), # chocolatemilk vs. placebo
       rq2 = c(1, -1, 0)) # chocolatemilk vs. dates
)
```

Power of each of the two contrasts:

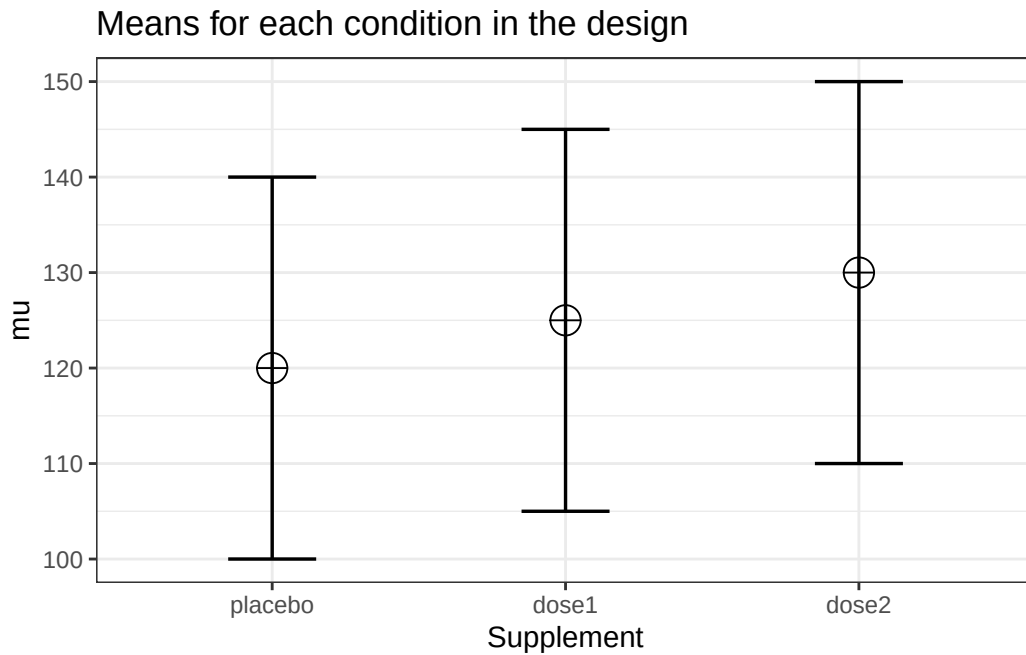
```
emmeans_power(contrast)
```

	contrast	power	partial_eta_squared	cohen_f	non centrality
1	rq1	64.49287	0.08982036	0.3141404	5.625
2	rq2	12.15715	0.01084599	0.1047135	0.625

Now suppose that we are interested in the following RQ:

- *Does the consumption of any dose of caffeine affects power output during a time trial in comparison to a placebo?*

```
design_study <- ANOVA_design(
  design = "3b",
  n = 40,
  mu = c(120, 125, 130),
  sd = 20, # common standard deviation
  labelnames = c("Supplement",
                 "placebo", "dose1", "dose2"),
  plot = TRUE
)
```

```
res <- ANOVA_exact(  
  design_study,  
  alpha_level = 0.05,  
  emm = TRUE  
)
```

```
comparisons <- emmeans(  
  res$emmeans$emmeans,  
  specs = ~ Supplement  
)
```

```
custom_contrast <- contrast(  
  comparisons,  
  list(c1 = c(1, 0, -1), # dose1 vs. placebo  
       c2 = c(0, 1, -1)) # dose2 vs. placebo  
)
```

Power for each of the two contrasts:

```
emmeans_power(  
  custom_contrast,  
  alpha_level = 0.05/2  
)
```

	contrast	power	partial_eta_squared	cohen_f	non centrality
1	c1	12.85260	0.01057082	0.1033623	1.25
2	c2	48.82994	0.04098361	0.2067246	5.00

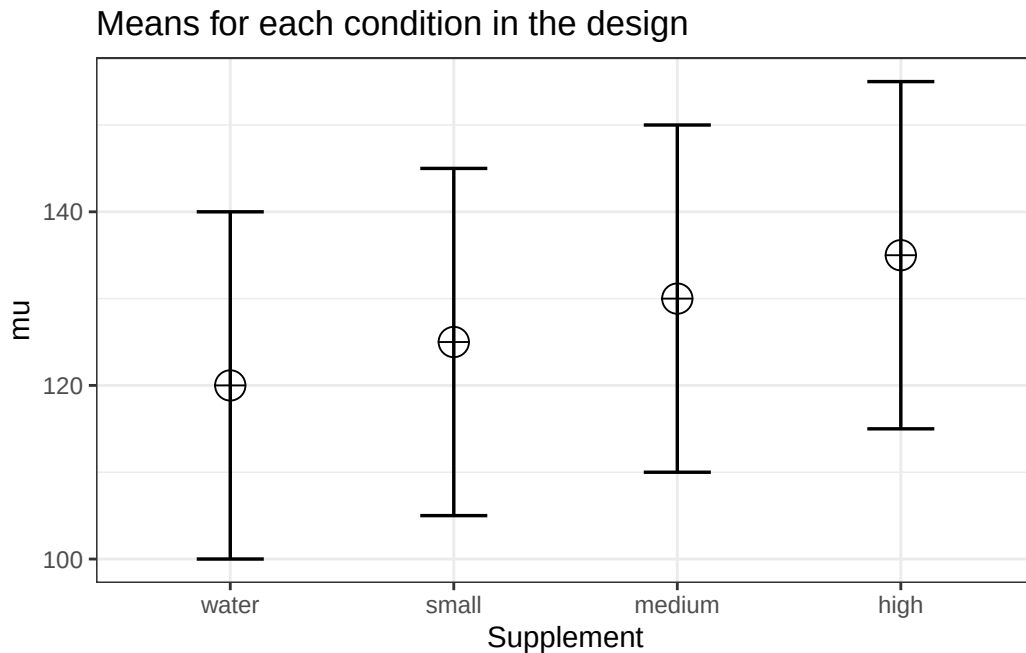
We adjust α (i.e., α divided by the number of comparisons) because we are testing a **disjunction hypothesis**, meaning we will consider the hypothesis supported if any dose significantly affects power output compared to the placebo.

Within-subject ANOVA

Suppose we are interested in testing the following research question:

- *Does increasing doses of caffeine produce a dose-response effect on power output during a time trial, compared with water (placebo)?*

```
design_result <- ANOVA_design(
  design = "4w",
  n = 30,
  mu = c(120, 125, 130, 135),
  sd = 20, # common standard deviation
  r = 0.7, # common correlation
  labelnames = c("Supplement",
                 "water", "small", "medium", "high"),
  plot = TRUE)
```



```
res <- ANOVA_exact(
  design_result,
  alpha_level = 0.05,
  emm = TRUE
)
```

```
comparisons <- emmeans(
  res$emmeans$emmeans,
  specs = ~ Supplement
)
```

We need to assign a set of contrasts to each condition based on the expected pattern of means. The contrast `linear_trend = c(-2, -1, 1, 2)` assigns weights to ordered groups to test for a linear increase or decrease across them; the coefficients sum to zero and give progressively larger positive weights to higher levels and negative weights to lower levels.

```
custom_contrast <- contrast(
  comparisons,
  list(linear_trend = c(-2, -1, 1, 2))
)
```

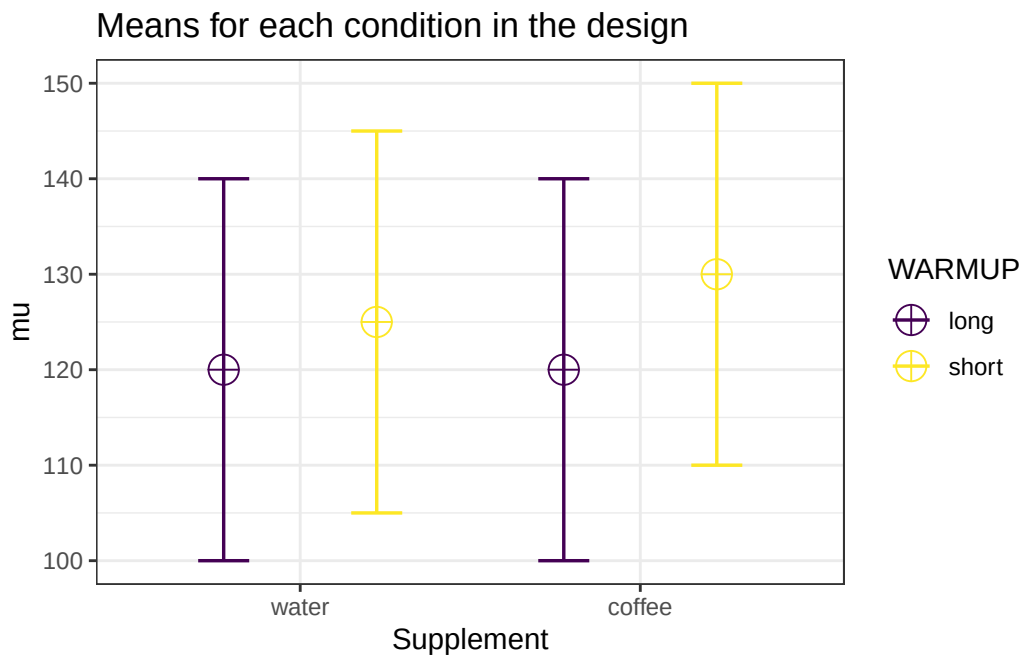
Power of the contrast test:

```
emmeans_power(custom_contrast)
```

	contrast	power	partial_eta_squared	cohen_f	non centrality
1	linear_trend	99.96419	0.5136268	1.027635	30.625

2x2 within-subject ANOVA

```
design_result <- ANOVA_design(  
  design = "2w*2w",  
  n = 80,  
  mu = c(120, 125, 120, 130),  
  sd = 20, # common standard deviation  
  r = 0.8, # common correlation  
  labelnames = c("Supplement",  
                 "water", "coffee",  
                 "WARMUP",  
                 "long", "short"),  
  plot = TRUE)
```



ANOVA_exact

Using `Superpower::ANOVA_exact` to conduct a power analysis performs only a single computation based on the specified parameters, meaning it yields an exact analytic solution for power.

```
res <- ANOVA_exact(
  design_result,
  alpha_level = 0.05
)
```

Power for main effects and interaction effect:

```
res$main_results
```

	power	partial_eta_squared	cohen_f	non centrality
Supplement	69.48206	0.07331378	0.2812720	6.25
WARMUP	100.00000	0.41589649	0.8438159	56.25
Supplement:WARMUP	69.48206	0.07331378	0.2812720	6.25

Power for pairwise comparisons:

```
res$pc_results
```

	power
p_Supplement_water_WARMUP_long_Supplement_water_WARMUP_short	93.72723
p_Supplement_water_WARMUP_long_Supplement_coffee_WARMUP_long	5.00000
p_Supplement_water_WARMUP_long_Supplement_coffee_WARMUP_short	99.99997
p_Supplement_water_WARMUP_short_Supplement_coffee_WARMUP_long	93.72723
p_Supplement_water_WARMUP_short_Supplement_coffee_WARMUP_short	93.72723
p_Supplement_coffee_WARMUP_long_Supplement_coffee_WARMUP_short	99.99997

	effect_size
p_Supplement_water_WARMUP_long_Supplement_water_WARMUP_short	3.952847e-01
p_Supplement_water_WARMUP_long_Supplement_coffee_WARMUP_long	2.808667e-17
p_Supplement_water_WARMUP_long_Supplement_coffee_WARMUP_short	7.905694e-01
p_Supplement_water_WARMUP_short_Supplement_coffee_WARMUP_long	-3.952847e-01
p_Supplement_water_WARMUP_short_Supplement_coffee_WARMUP_short	3.952847e-01
p_Supplement_coffee_WARMUP_long_Supplement_coffee_WARMUP_short	7.905694e-01

`ANOVA_power`

Using `Superpower::ANOVA_power` to conduct a power analysis relies on Monte Carlo simulations, meaning it repeatedly generates many random datasets based on the specified design

parameters and estimates statistical power as the proportion of simulations in which the effect is detected (i.e., $p < \alpha$).

```
res_sim <- ANOVA_power(
  design_result,
  alpha_level = 0.05,
  nsims = 1000
)
```

Power for main effects and interaction effect:

```
res_sim$main_results
```

	power	effect_size
anova_Supplement	69.1	0.08427715
anova_WARMUP	100.0	0.41567728
anova_Supplement:WARMUP	69.0	0.08003715

Power for pairwise comparisons:

```
res_sim$pc_results
```

	power	effect_size
p_Supplement_water_WARMUP_long_Supplement_water_WARMUP_short	93.4	0.399412939
p_Supplement_water_WARMUP_long_Supplement_coffee_WARMUP_long	5.0	0.006387097
p_Supplement_water_WARMUP_long_Supplement_coffee_WARMUP_short	100.0	0.800887136
p_Supplement_water_WARMUP_short_Supplement_coffee_WARMUP_long	93.5	-0.392768972
p_Supplement_water_WARMUP_short_Supplement_coffee_WARMUP_short	93.4	0.396221200
p_Supplement_coffee_WARMUP_long_Supplement_coffee_WARMUP_short	100.0	0.792646923

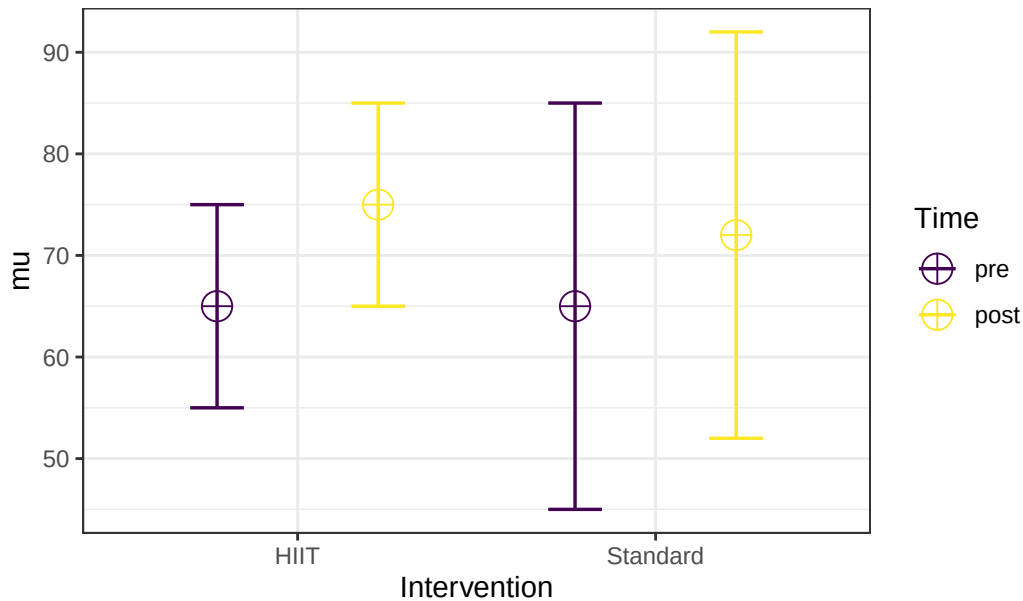
2x2 mixed ANOVA

Now suppose we are interested in the following RQ:

- Does a 4-week HIIT training (2 sessions per week) lead to changes in VO_{2max} at 6 weeks post-intervention?

```
design_result <- ANOVA_design(  
  design = "2b*2w",  
  n = 80,  
  mu = c(65, 75, 65, 72),  
  sd = c(10, 10, 20, 20), # assing different SDs  
  r = 0.8,                # common correlation  
  labelnames = c("Intervention",  
                 "HIIT", "Standard",  
                 "Time",  
                 "pre", "post"),  
  plot = TRUE)
```

Means for each condition in the design



Power and Effect sizes for ANOVA tests

	power	partial_eta_squared	cohen_f	non centrality
Intervention	9.6356	0.0025	0.0503	0.4
Time	100.0000	0.4225	0.8554	115.6

Intervention:Time	47.0512	0.0223	0.1509	3.6
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Power and Effect sizes for pairwise comparisons (t-tests)

	power
p_Intervention_HIIT_Time_pre_Intervention_HIIT_Time_post	100.00
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_pre	5.00
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_post	79.47
p_Intervention_HIIT_Time_post_Intervention_Standard_Time_pre	97.81
p_Intervention_HIIT_Time_post_Intervention_Standard_Time_post	22.23
p_Intervention_Standard_Time_pre_Intervention_Standard_Time_post	99.83

	effect_size
p_Intervention_HIIT_Time_pre_Intervention_HIIT_Time_post	1.58
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_pre	0.00
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_post	0.44
p_Intervention_HIIT_Time_post_Intervention_Standard_Time_pre	-0.63
p_Intervention_HIIT_Time_post_Intervention_Standard_Time_post	-0.19
p_Intervention_Standard_Time_pre_Intervention_Standard_Time_post	0.55

Power for main effects and interaction effects:

```
res$main_results
```

	power	partial_eta_squared	cohen_f	non centrality
Intervention	9.635632	0.002525253	0.05031546	0.4
Time	100.000000	0.422514620	0.85536283	115.6
Intervention:Time	47.051166	0.022277228	0.15094638	3.6

Power for the comparison between interventions at week 6:

```
res$pc_results[5,]
```

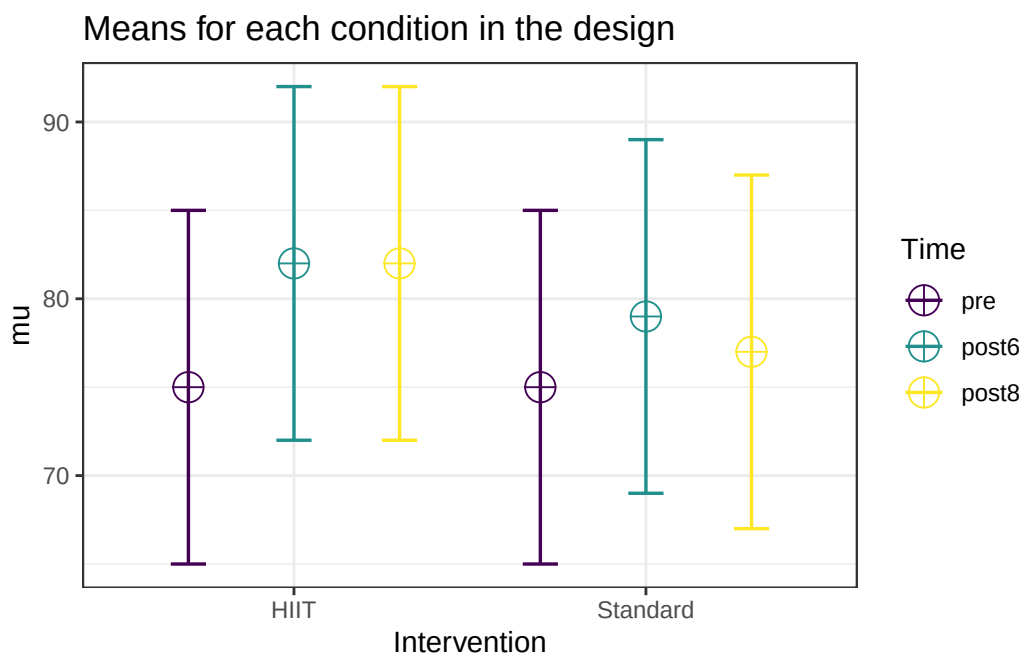
	power
p_Intervention_HIIT_Time_post_Intervention_Standard_Time_post	22.22751

	effect_size
p_Intervention_HIIT_Time_post_Intervention_Standard_Time_post	-0.1897367

Now suppose we are interested in answering the following RQ:

- Does a 4-week HIIT training (2 sessions per week) lead to changes in VO_{2max} at 6 and 8 weeks post-intervention?

```
design_result <- ANOVA_design(
  design = "2b*3w",
  n = 80,
  mu = c(75, 82, 82, 75, 79, 77),
  sd = 10,
  r = 0.8,                      # common correlation
  labelnames = c("Intervention",
                 "HIIT", "Standard",
                 "Time",
                 "pre", "post6", "post8"),
  plot = TRUE)
```



Power and Effect sizes for ANOVA tests

	power	partial_eta_squared	cohen_f	non centrality
Intervention	43.6787	0.0203	0.1441	3.2821
Time	100.0000	0.3029	0.6592	137.3333
Intervention:Time	99.6569	0.0742	0.2831	25.3333

Power and Effect sizes for pairwise comparisons (t-tests)

	power
p_Intervention_HIIT_Time_pre_Intervention_HIIT_Time_post6	100.00
p_Intervention_HIIT_Time_pre_Intervention_HIIT_Time_post8	100.00
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_pre	5.00
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_post6	71.04
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_post8	24.18
p_Intervention_HIIT_Time_post6_Intervention_HIIT_Time_post8	5.00
p_Intervention_HIIT_Time_post6_Intervention_Standard_Time_pre	99.27
p_Intervention_HIIT_Time_post6_Intervention_Standard_Time_post6	47.05
p_Intervention_HIIT_Time_post6_Intervention_Standard_Time_post8	88.16
p_Intervention_HIIT_Time_post8_Intervention_Standard_Time_pre	99.27
p_Intervention_HIIT_Time_post8_Intervention_Standard_Time_post6	47.05
p_Intervention_HIIT_Time_post8_Intervention_Standard_Time_post8	88.16
p_Intervention_Standard_Time_pre_Intervention_Standard_Time_post6	99.99
p_Intervention_Standard_Time_pre_Intervention_Standard_Time_post8	79.78
p_Intervention_Standard_Time_post6_Intervention_Standard_Time_post8	79.78
	effect_size
p_Intervention_HIIT_Time_pre_Intervention_HIIT_Time_post6	1.11
p_Intervention_HIIT_Time_pre_Intervention_HIIT_Time_post8	1.11
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_pre	0.00
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_post6	0.40
p_Intervention_HIIT_Time_pre_Intervention_Standard_Time_post8	0.20
p_Intervention_HIIT_Time_post6_Intervention_HIIT_Time_post8	0.00
p_Intervention_HIIT_Time_post6_Intervention_Standard_Time_pre	-0.70
p_Intervention_HIIT_Time_post6_Intervention_Standard_Time_post6	-0.30
p_Intervention_HIIT_Time_post6_Intervention_Standard_Time_post8	-0.50
p_Intervention_HIIT_Time_post8_Intervention_Standard_Time_pre	-0.70
p_Intervention_HIIT_Time_post8_Intervention_Standard_Time_post6	-0.30
p_Intervention_HIIT_Time_post8_Intervention_Standard_Time_post8	-0.50
p_Intervention_Standard_Time_pre_Intervention_Standard_Time_post6	0.63
p_Intervention_Standard_Time_pre_Intervention_Standard_Time_post8	0.32
p_Intervention_Standard_Time_post6_Intervention_Standard_Time_post8	-0.32

Power and Effect sizes for estimated marginal means

	contrast	power
1	Intervention_HIIT Time_post6 - Intervention_Standard Time_post6	47.05
2	Intervention_HIIT Time_post6 - Intervention_HIIT Time_post8	5.00
3	Intervention_HIIT Time_post6 - Intervention_Standard Time_post8	88.16
4	Intervention_HIIT Time_post6 - Intervention_HIIT Time_pre	100.00
5	Intervention_HIIT Time_post6 - Intervention_Standard Time_pre	99.27
6	Intervention_Standard Time_post6 - Intervention_HIIT Time_post8	47.05
7	Intervention_Standard Time_post6 - Intervention_Standard Time_post8	80.27
8	Intervention_Standard Time_post6 - Intervention_HIIT Time_pre	71.04

9	Intervention_Standard Time_post6 - Intervention_Standard Time_pre	99.99
10	Intervention_HIIT Time_post8 - Intervention_Standard Time_post8	88.16
11	Intervention_HIIT Time_post8 - Intervention_HIIT Time_pre	100.00
12	Intervention_HIIT Time_post8 - Intervention_Standard Time_pre	99.27
13	Intervention_Standard Time_post8 - Intervention_HIIT Time_pre	24.18
14	Intervention_Standard Time_post8 - Intervention_Standard Time_pre	80.27
15	Intervention_HIIT Time_pre - Intervention_Standard Time_pre	5.00
	partial_eta_squared	cohen_f
1	0.0223	0.1509
2	0.0000	0.0000
3	0.0595	0.2516
4	0.3828	0.7876
5	0.1104	0.3522
6	0.0223	0.1509
7	0.0482	0.2250
8	0.0389	0.2013
9	0.1684	0.4500
10	0.0595	0.2516
11	0.3828	0.7876
12	0.1104	0.3522
13	0.0100	0.1006
14	0.0482	0.2250
15	0.0000	0.0000

```
emm_results <- res$emmeans$emmeans
```

Compare HIIT vs Standard at each time point (post6, post8, etc.):

```
pairwise_results <- emmeans(
  emm_results,
  pairwise ~ Intervention | Time
)
```

Power of each comparison:

```
emmeans_power(pairwise_results$contrasts)
```

	contrast	Time	power
1	Intervention_HIIT - Intervention_Standard	Time_post6	47.05117
2	Intervention_HIIT - Intervention_Standard	Time_post8	88.16025
3	Intervention_HIIT - Intervention_Standard	Time_pre	5.00000

	partial_eta_squared	cohen_f	non centrality
1	2.227723e-02	1.509464e-01	3.600000e+00
2	5.952381e-02	2.515773e-01	1.000000e+01
3	1.192690e-33	3.453535e-17	1.884451e-31

Linear model regression with one single dichotomous predictor

The general equation for a linear model is:

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + \epsilon$$

We can use this model to simulate data.

Suppose an article investigated the effect of endurance training on VO_{2max} using a two independent-groups study design.

```
set.seed(123)

n <- 60 # sample size

# Simulate data
df <- data.frame(
  exp = c(rep(0, n/2), rep(1, n/2)),
  outcome = c(rnorm(n/2, 58, 8),
              rnorm(n/2, 60, 8))
)

# Fit the model
(model <- lm(outcome ~ exp, df) |> tidy())
```

```
# A tibble: 2 x 5
  term      estimate std.error statistic  p.value
<chr>      <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)  57.6       1.33      43.3 7.26e-46
2 exp          3.80       1.88       2.02 4.79e- 2
```

In a RCT, the intervention effect corresponds to the estimate of the slope. The standardized effect size estimate can be expressed as Cohen's d , which is the mean difference divided by the standard deviation (SD):

```
(es <- model$estimate[2]/sd(df$outcome))
```

```
[1] 0.508754
```

This effect size estimate can be used to conduct an a priori power analysis:

```
pwr.t.test(  
  power = 0.9,  
  d = es,  
  sig.level = 0.05,  
  type = "two.sample",  
  alternative = "two.sided"  
)
```

Two-sample t test power calculation

```
      n = 82.16381  
      d = 0.508754  
sig.level = 0.05  
  power = 0.9  
alternative = two.sided
```

NOTE: n is number in *each* group

We could use the same model and estimate of the intervention effect to conduct a simulation-based power analysis:

```
set.seed(123)  
  
n <- 160 # total sample size  
exp <- c(rep(0, n/2), rep(1, n/2)) # assign participants to 0 or 1  
sd <- sd(df$outcome) # calculate SD  
  
# Write down the equation  
outcome <- 55 + 4*exp + rnorm(n, mean=0, sd=sd)  
  
# combine all variables in one data frame  
df <- data.frame(exp, outcome)  
  
head(df)
```

	exp	outcome
1	0	50.80978
2	0	53.27915
3	0	66.65318
4	0	55.52713
5	0	55.96658
6	0	67.82214

Fit the ANOVA model:

```
lm(outcome ~ exp, df) |>
  tidy()
```

```
# A tibble: 2 x 5
  term          estimate std.error statistic  p.value
<chr>         <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    55.1      0.782      70.5 2.45e-121
2 exp             3.49      1.11       3.16 1.91e- 3
```

Due to sampling variability, the estimate of the intervention effect will vary from sample to sample. We can perform a simulation-based power analysis to account for this uncertainty (Monte Carlo simulation).

```
# Create a simulation function
sim1 <- function(effect, n){

  exp <- c(rep(0, n/2), rep(1, n/2))

  #Simulate the data
  outcome <- 55 - effect*exp + rnorm(n, mean=0, sd=sd)

  # Fit ANOVA model
  res <- tidy(lm(outcome ~ exp))

  # Extract the p-value
  p_value <- res$p.value[2]
}

# Run the simulation function
p_values <- replicate(1000, sim1(effect = 5, n = 100))
```

```
# Count how many p-values are smaller than the alpha level
mean(p_values < 0.05)
```

```
[1] 0.913
```

In a Monte Carlo simulation, the power of the test is nothing else than the number of p -values that fall below the alpha level.

Linear model regression with one single dichotomous predictor and a baseline covariate

In the morning we saw that including a baseline covariate can boost power. To simulate data for a factorial design, we can use the **faux** package which allows to easily simulate correlated data.

```
set.seed(123)

n <- 80
sd <- 7.19
r <- 0.6

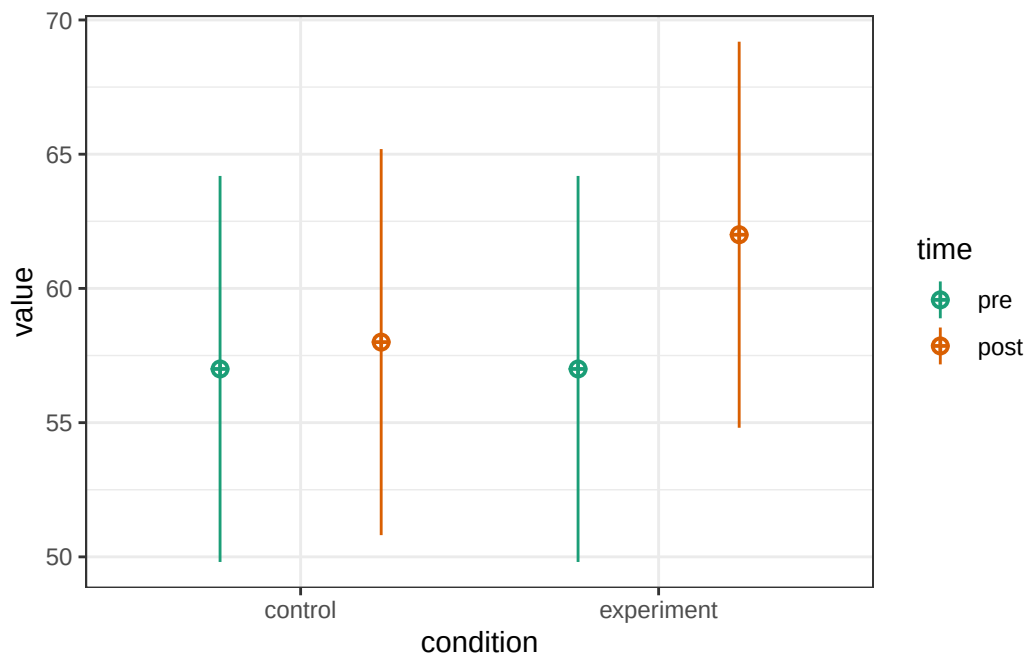
# One between-subject factor with two levels
between <- list(condition = c(con = "control", exp = "experiment"))

# One within-subject factor with two levels
within <- list(time = c("pre", "post"))

# Set means
mu <- data.frame(
  con = c(57, 58),
  exp = c(57, 62),
  row.names = within$time
)

# Simulate data
df2 <- sim_design(
  within,
  between,
  n = n,
  mu = mu,
  sd = sd,
```

```
r = r
)
```



```
# Fit ANCOVA model
(model <- lm(post ~ condition + pre, df2) |>
  tidy())
```

```
# A tibble: 3 x 5
  term          estimate std.error statistic  p.value
<chr>         <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    21.7       3.33        6.52 9.26e-10
2 conditionexp    2.68       0.843        3.18 1.77e- 3
3 pre            0.648      0.0573       11.3 4.74e-22
```

```
sim2 <- function(n, r){
  # One between-subject factor with two levels
  between <- list(condition = c(con = "control", exp = "experiment"))

  # One within-subject factor with two levels
  within <- list(time = c("pre", "post"))
```



```

# Set means
mu <- data.frame(
  con = c(57, 58),
  exp = c(58, 62),
  row.names = within$time
)

# Simulate data
df2 <- sim_design(
  within,
  between,
  n = n,
  mu = mu,
  sd = 8,
  r = r,
  plot = FALSE
)

# Fit ANCOVA model
model <- lm(post ~ condition + pre, df2) |>
  tidy()

# Extract p-value for 'condition'
p_value <- model$p.value[2]
}

# Run the simulation function
p_values <- replicate(1000, sim2(n = 80, r = 0.6))

# Count how many p-values are smaller than the alpha level
mean(p_values < 0.05)

```

```
[1] 0.912
```

Now let's rerun the simulation but without including the baseline covariate and let's compare the statistical power of the design against with the power achieved by the ANCOVA model.

```

sim3 <- function(n, r){

# One between-subject factor with two levels
between <- list(condition = c(con = "control", exp = "experiment"))

```

```

# One within-subject factor with two levels
within <- list(time = c("pre", "post"))

# Set means
mu <- data.frame(
  con = c(57, 58),
  exp = c(58, 62),
  row.names = within$time
)

# Simulate data
df2 <- sim_design(
  within,
  between,
  n = n,
  mu = mu,
  sd = 8,
  r = r,
  plot = FALSE
)

# Fit ANCOVA model
model <- lm(post ~ condition, df2) |>
  tidy()

# Extract p-value for 'condition'
p_value <- model$p.value[2]
}

# Run the simulation function
p_values <- replicate(1000, sim3(n = 80, r = 0.6))

# Count how many p-values are smaller than the alpha level
mean(p_values < 0.05)

```

```
[1] 0.873
```

Instead of relying on trial and error to determine the sample size, we can use `map_dbl()` to run simulations across a range of sample sizes and efficiently estimate the corresponding power for each.

```

# sequence of sample sizes
n <- seq(30, 120, 10)

# Compute estimated power for each n
power_estimates <- map_dbl(n, function(nsize) {
  p_values <- replicate(1000, sim2(n = nsize, r = 0.6))
  mean(p_values < 0.05)
})

# Combine results into a data frame
power_df <- data.frame(
  n = n,
  power = power_estimates
)

power_df

```

	n	power
1	30	0.524
2	40	0.669
3	50	0.757
4	60	0.797
5	70	0.877
6	80	0.887
7	90	0.944
8	100	0.950
9	110	0.973
10	120	0.985

Difference in difference approach to estimate interaction effects

One simple approach to estimate an effect size for an interaction effect is based on the difference-in-difference approach (Sommet et al. 2023); Langenberg et al. (2023)].

```
knitr::include_graphics("figures/simple_effect.png")
```

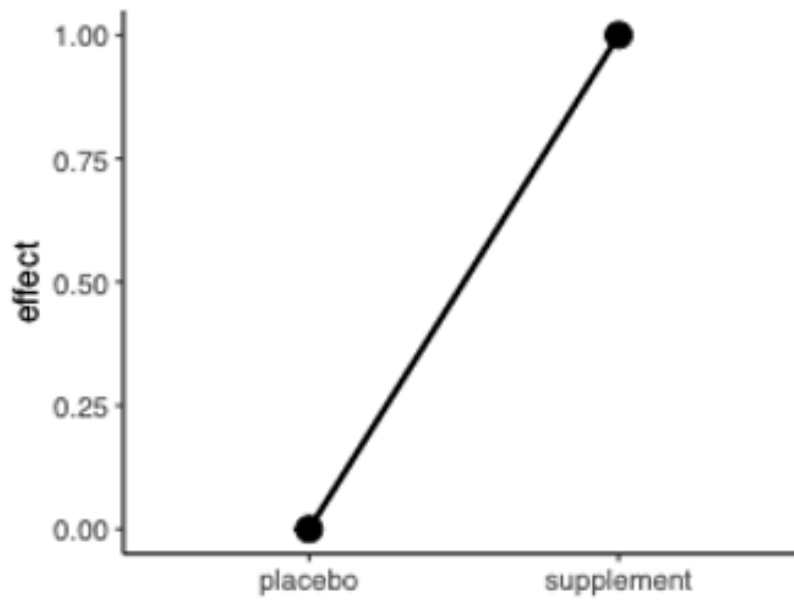


Figure 1

In Figure 1, we would use Equation 1 to estimate a standardised effect size (Cohen's d_s) for a two-group design.

$$d_s = \frac{(E - C)}{SD_{pooled}} \quad (1)$$

Using Equation 1, we would obtain a Cohen's d_s of:

```
m1 <- 1 # intervention group mean
m2 <- 0 # control group mean
sd <- 2 # pooled standard deviation

(d <- (m1 - m2)/sd)
```

```
[1] 0.5
```

A common issue in sports science is the uncritical adoption of effect sizes from previous studies based on simpler designs, which may not align with the factorial structure of the current study when conducting a power analysis for an interaction effect. Whether Cohen's d_s is an

appropriate estimate of the interaction effect depends on the underlying pattern of means, that is, the type of interaction.

```
knitr::include_graphics("figures/interaction.png")
```

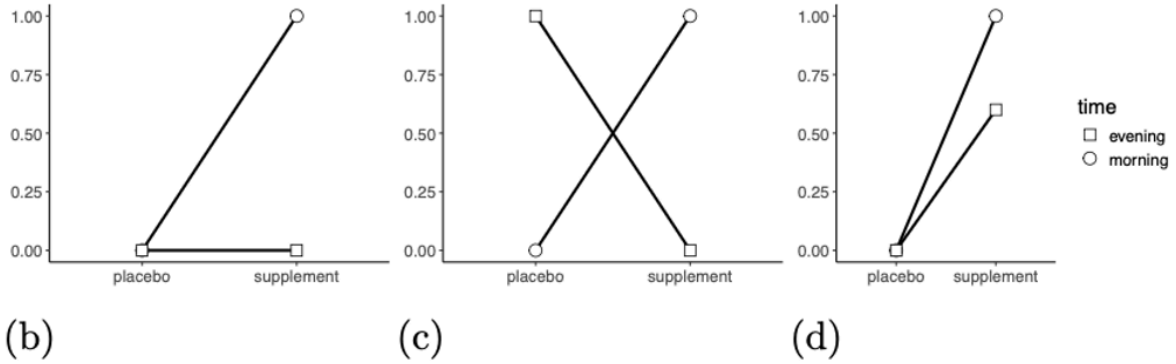


Figure 2: b) Fully attenuated interaction, c) crossover interaction, d) attenuated interaction

Using Equation 2, we can estimate the interaction effect based on the pattern of means observed in a 2×2 between-subjects design. This effect size is calculated as a “difference in differences” as follows (Sommet et al. 2023):

$$d_{int} = \frac{(E_2 - C_2) - (E_1 - C_1)}{(2 * SD_{pooled})} \quad (2)$$

For Figure 2 b, the interaction effect (fully attenuated interaction) would be:

```
E1 <- 0
C1 <- 0
E2 <- 1
C2 <- 0
sd <- 2
((E2 - C2) - (E1 - C1)) / (2*sd)
```

```
[1] 0.25
```

For Figure 2 c, the interaction effect (crossover interaction) would be:

```
E1 <- 0
C1 <- 1
E2 <- 1
C2 <- 0
sd <- 2
((E2 - C2) - (E1 - C1)) / (2*sd)
```

```
[1] 0.5
```

For Figure 2 d, the interaction effect (attenuated interaction) would be:

```
E1 <- 0.6
C1 <- 0
E2 <- 1
C2 <- 0
sd <- 2
((E2 - C2) - (E1 - C1)) / (2*sd)
```

```
[1] 0.1
```

Once the interaction effect size has been estimated, we can plug in the estimated Cohen's d_{int} into the statistical software. For example:

```
sample <- pwr.t.test(d = 0.1, power = 0.9, sig.level = 0.05, type = "two.sample", alternative = "two.sided")
sample$n
```

```
[1] 2102.445
```

We can also use the same in difference-in-difference approach for factorial within-subject designs (Langenberg et al. 2023). The only difference is that we need to specify the

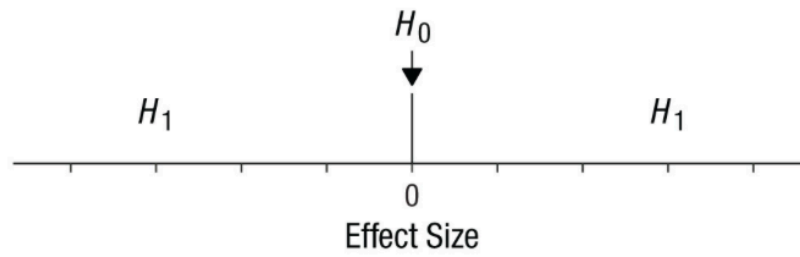
Equivalence tests

Sports scientists typically aim to reject the null hypothesis of zero (i.e., $H_0 = 0$) to conclude that two interventions differ, or that one is superior or inferior to a standard intervention. However, in some cases, researchers are instead interested in determining whether two interventions are equivalent (Lakens, Scheel, and Isager 2018; Mazzolari et al. 2022). In such situations, an equivalence test must be conducted.

```
knitr::include_graphics("figures/types_hypothesis.png")
```

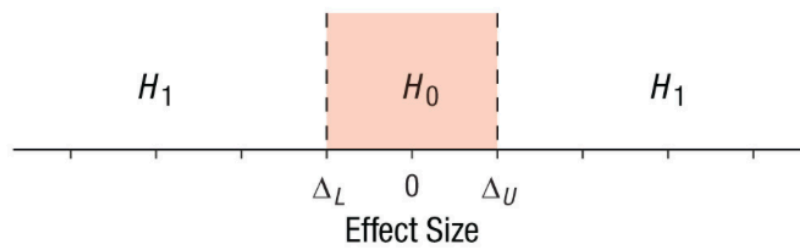
a

Classic Null-Hypothesis Significance Test (Two Sided)



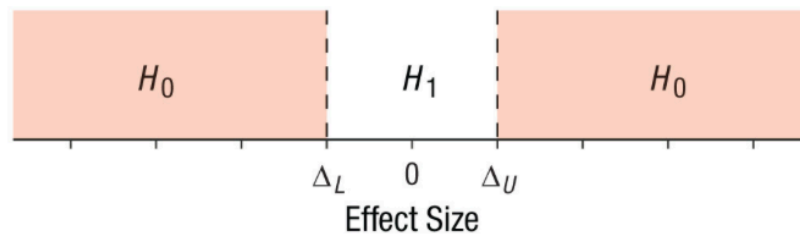
b

Minimal-Effects Test



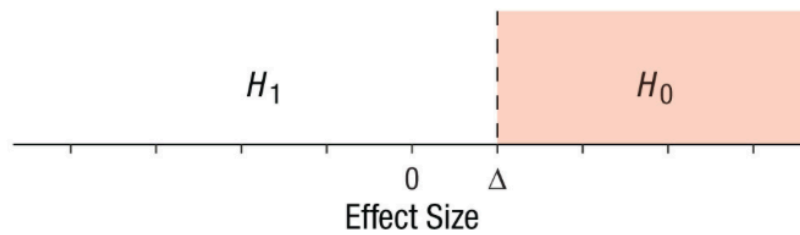
c

Equivalence Test



d

Inferiority Test



Suppose we are interested in answering the following RQ:

- *Is the effect of taking a nap comparable to cryotherapy on force loss after a football match, as measured during an isokinetic knee test?*

```
n <- 30          # sample size
d <- 0.1         # expected mean difference
sd <- 1          # expected standard deviation
low <- -0.3      # lower bound of the region of equivalence
high <- 0.3      # upper bound of the region of equivalence
type <- "two_sample" # type of test ("one_sample", "two_sample")

power_t_TOST(
  n = n,
  delta = d,
  sd = sd,
  low_eqbound = low,
  high_eqbound = high,
  alpha = 0.05,
  type = type
)
```

TOST power calculation

```
power = 0.04706003
beta = 0.95294
alpha = 0.05
n = 30
delta = 0.1
sd = 1
bounds = -0.3, 0.3
```

! Important

The closer the effect size (d) is to the equivalence bounds, the lower the statistical power. Both the effect size (d) and the equivalence bounds should be preregistered, and a clear rationale should be provided to avoid using overly wide equivalence margins.

```

# Create a sequence of sample sizes
n_seq <- seq(50, 250, 25)

# Compute power for each sample size
power_estimates <- map_dbl(n_seq, function(nsize) {
  power_t_TOST(
    n = nsize,
    delta = d,
    sd = sd,
    low_eqbound = low,
    high_eqbound = high,
    alpha = 0.05,
    type = type
  )$power
})

# Combine into data frame
power_df <- data.frame(
  n = n_seq,
  power = power_estimates
)

power_df

```

	n	power
1	50	0.2751168
2	75	0.4914904
3	100	0.6236512
4	125	0.7161677
5	150	0.7856387
6	175	0.8389128
7	200	0.8797574
8	225	0.9108524
9	250	0.9343254

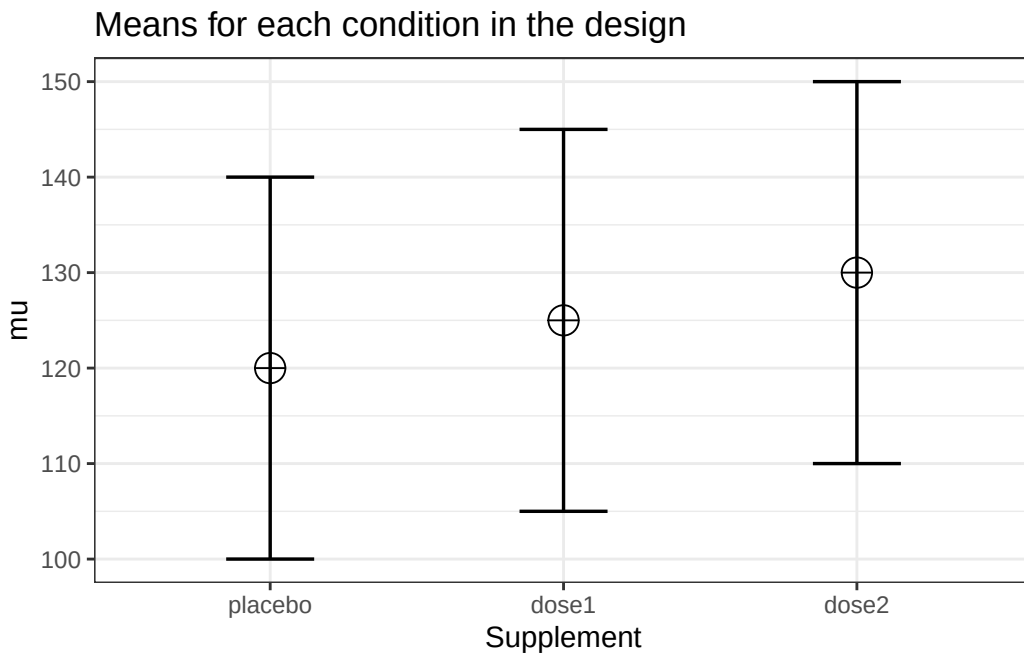
We can also perform equivalence tests using **Superpower**. Now suppose we are interested in the following RQ:

- *Is the difference in power output between the two doses during a time trial smaller than 0.1?*

```

design_study <- ANOVA_design(
  design = "3b",
  n = 40,
  mu = c(120, 125, 130),
  sd = 20, # common standard deviation
  labelnames = c("Supplement",
                 "placebo", "dose1", "dose2"),
  plot = TRUE
)

```



```

res <- ANOVA_exact(
  design_study,
  alpha_level = 0.05,
  emm = TRUE
)

```

Power and Effect sizes for ANOVA tests

	power	partial_eta_squared	cohen_f	non centrality
Supplement	49.2898	0.041	0.2067	5

Power and Effect sizes for pairwise comparisons (t-tests)

	power	effect_size
--	-------	-------------

p_Supplement_placebo_Supplement_dose1	19.72	0.25
p_Supplement_placebo_Supplement_dose2	59.81	0.50
p_Supplement_dose1_Supplement_dose2	19.72	0.25

Power and Effect sizes for estimated marginal means

	contrast	power	partial_eta_squared	cohen_f
1	Supplement_dose1 - Supplement_dose2	19.84	0.0106	0.1034
2	Supplement_dose1 - Supplement_placebo	19.84	0.0106	0.1034
3	Supplement_dose2 - Supplement_placebo	60.17	0.0410	0.2067

	non_centralality
1	1.25
2	1.25
3	5.00

```
comparisons <- emmeans(
  res$emmeans$emmeans,
  specs = ~ Supplement
)

custom_contrast <- contrast(
  comparisons,
  list(c3 = c(1, -1, 0)) # dose1 vs. dose2
)
```

```
emmeans_power(
  custom_contrast,
  side = "equivalence",
  delta = 0.1
)
```

	contrast	power	partial_eta_squared	cohen_f	non_centralality
1	c3	19.84334	0.01057082	0.1033623	1.25

Generalized Linear model regression with one dichotomous outcome

Generalized linear models (GLMs) are a class of statistical models that include more complex models such as logistic regression and Poisson regression. In this context, we will focus on conducting a simulation-based power analysis for logistic regression. Recall that logistic regression is appropriate when the outcome variable is dichotomous, while the predictors can be either continuous or categorical.

```

set.seed(111)

dep_exp <- 0.3 # expected prob. of depression in EXP group
dep_con <- 0.5 # expected prob. of depression in CON group
n <- 100      # sample size

# Create a dichotomous outcome (1 = depression, 0 = no depression) for the experimental group
exp <- data.frame(
  intervention = "exp",
  depression = sample(
    c(1, 0),
    size = n,
    replace = TRUE,
    prob = c(dep_exp, 1-dep_exp))
)

# Create a dichotomous outcome (1 = depression, 0 = no depression) for the control group
con <- data.frame(
  intervention = "con",
  depression = sample(c(1, 0), size = n, replace = TRUE, prob = c(dep_con, 1-dep_con))
)

# Combine
df <- bind_rows(exp, con)

# Convert to factors
df$intervention <- factor(df$intervention)
df$depression <- factor(df$depression)

# Contingency table
table(df)

```

```

      depression
intervention 0  1
con      43 57
exp      75 25

```

As shown in the frequency table, the probability of depression (1) is lower in the experimental group and closely matches the true expected probabilities (40% in the experimental group and 50% in the control group).

```
(res_glm <- glm(depression ~ intervention, df, family = "binomial") |>
  tidy())
```

A tibble: 2 x 5

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	0.282	0.202	1.40	0.163
2	interventionexp	-1.38	0.307	-4.50	0.00000681

Convert to odds ratio

```
exp(res_glm$estimate[2])
```

```
[1] 0.251462
```

```
# Create a simulation function
sim5 <- function(dep_exp, dep_con, n){

  # Experimental group
exp <- data.frame(
  intervention = "exp",
  depression = sample(c(1, 0), size = n, replace = TRUE, prob = c(dep_exp, 1-dep_exp))
)

# Control group
con <- data.frame(
  intervention = "con",
  depression = sample(c(1, 0), size = n, replace = TRUE, prob = c(dep_con, 1-dep_con))
)

# Combine
df <- bind_rows(exp, con)
# Fit logistic model
res <- tidy(glm(depression ~ intervention, df, family = "binomial"))
# Extract the p-value
p_value <- res$p.value[2]
}

# Run the simulation function
p_values <- replicate(
  1000,
```

```

sim5(dep_exp, dep_con, n = 100)
)

# Count how many p-values are smaller than the alpha level
mean(p_values < 0.05)

```

```
[1] 0.815
```

```

LRcat(
  p0 = dep_con,
  p1 = dep_exp,
  prop = .50,
  alpha = .05,
  power = 0.83
)

```

- Caldwell, Aaron R., Daniël Lakens, Chelsea M. Parlett-Pelleriti, Guy Prochilo, and Frederik Aust. 2022. *Power Analysis with Superpower*. <https://aaroncaldwell.us/SuperpowerBook/index.html#preface>.
- Lakens, Daniël, Anne M. Scheel, and Peder M. Isager. 2018. “Equivalence Testing for Psychological Research: A Tutorial.” *Advances in Methods and Practices in Psychological Science* 1 (2): 259–69. <https://doi.org/10.1177/2515245918770963>.
- Langenberg, Benedikt, Markus Janczyk, Valentin Koob, Reinhold Kliegl, and Axel Mayer. 2023. “A Tutorial on Using the Paired t Test for Power Calculations in Repeated Measures ANOVA with Interactions.” *Behavior Research Methods* 55 (5): 2467–84. <https://doi.org/10.3758/s13428-022-01902-8>.
- Mazzolari, Raffaele, Simone Porcelli, David J. Bishop, and Daniël Lakens. 2022. “Myths and methodologies: The use of equivalence and non-inferiority tests for interventional studies in exercise physiology and sport science.” *Experimental Physiology* 107 (3): 201–12. <https://doi.org/10.1113/EP090171>.
- Sommet, Nicolas, David L. Weissman, Nicolas Cheutin, and Andrew J. Elliot. 2023. “How Many Participants Do I Need to Test an Interaction? Conducting an Appropriate Power Analysis and Achieving Sufficient Power to Detect an Interaction.” *Advances in Methods and Practices in Psychological Science* 6 (3): 25152459231178728. <https://doi.org/10.1177/25152459231178728>.